

Smart Farm Work Monitoring & Management Platform

Product Requirements Document (PRD)

1. Project Summary

Project Name

AgriFlow – Smart Farm Task Monitoring Platform

Estimated Project Value

₹80 Lakh Enterprise Software Project

Purpose

Build a secure, enterprise-level agriculture management platform where:

- Customers (Land Owners) can monitor farming activities remotely.
 - Workers/Agents upload daily farming work with photos/videos/PDF reports.
 - Owners assign and track farming tasks.
 - Admin controls all users, land IDs, and system security.
 - System supports multi-tenant architecture and real-time monitoring.
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2. Core Objectives

Business Objectives

- Digitize farming workflow management.
- Improve transparency between land owner and workers.
- Provide proof-based farming reports using media uploads.
- Reduce communication gaps.
- Create scalable enterprise agricultural SaaS platform.

Technical Objectives

- Fast response time (< 3 seconds)

- Highly secure authentication system
 - Mobile + desktop responsive UI
 - Scalable backend architecture
 - Multi-tenant support
 - Enterprise-grade file handling
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3. Technology Stack

Frontend

Framework

- React JS (Latest Stable Version)

UI Requirements

- Responsive Design
- Mobile + Tablet + Desktop support
- Premium Agri-based Theme

Design Theme

Colors:

- Plant Green
- Sky Blue
- Transparent Water Effects
- Modern Glassmorphism UI

Frontend Performance

- Fast page loading
 - Lazy loading
 - Image optimization
 - Skeleton loaders
 - CDN support
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Backend

Framework

- Python FastAPI

Requirements

- High-speed API responses
 - Async processing
 - Error-free business logic
 - JWT Authentication
 - RBAC (Role Based Access Control)
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Database

Database Engine

- PostgreSQL (Latest Version)

Local Database Configuration

- PostgreSQL Local Server
- Password: 271527

Database Requirements

- Retrieval speed:
- Maximum: 3 seconds
- Target: Less than 1 second

DB Features

- Indexing
 - Query optimization
 - Multi-tenant schema support
 - Backup & recovery
 - Transaction safety
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4. User Roles

4.1 CUSTOMER (Land Owner)

Permissions

Customer Can:

- Login securely

- View assigned lands
- View farming process updates
- View:
- Photos
- Videos
- PDFs
- Task timelines
- Download files
- View process date/time
- Raise comments
- Submit complaints
- Communicate with owner

Restrictions

Customer Cannot:

- Edit tasks
- Delete reports
- Upload farming reports
- Modify workflow

4.2 AGENT / WORKER

Permissions

Worker Can:

- Upload today's farming work
- Upload:
- Multiple photos
- Multiple videos
- PDF reports
- Update task progress
- Add farming notes
- Communicate with:
- Customer
- Owner
- Mark work status:
- Started
- In Progress
- Completed
- Add geo-location (optional future feature)

Upload Requirements

- Drag & Drop upload
 - Multi-file upload
 - Compression support
 - Auto-thumbnail generation
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4.3 OWNER

Permissions

Owner Can:

- Create farming tasks
- Assign tasks to workers
- Track task completion
- Search using Land ID
- View:
 - Customer details
 - Worker reports
 - Work timelines
 - Pending tasks
 - Completed tasks
 - Monitor land-wise activity
 - View agent performance reports

Dashboard Features

- Task analytics
 - Worker productivity
 - Land progress charts
 - Media gallery
 - Complaint management
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4.4 ADMIN

Super Admin Permissions

Admin Controls:

- Entire platform
- User creation

- User deletion
- Role management
- Land ID generation
- Login system
- Security management
- Audit logs

Admin Can Create:

- Customer accounts
- Owner accounts
- Agent accounts
- Admin accounts

Mandatory Fields During User Creation

- Username
- Email ID
- Role
- Land ID
- Password

Search Features

Admin can search using:

- Land ID
- Name
- Email ID

Activity Monitoring

Admin can monitor:

- Login time
- Logout time
- Agent activity
- System usage
- Failed login attempts

5. Functional Requirements

5.1 Authentication System

Security Requirements

- JWT Authentication
- Refresh Tokens
- Password Hashing (bcrypt/argon2)
- HTTPS Only
- Session timeout
- Device tracking
- Role-based authorization
- Login audit logs
- IP logging

Login Features

- Secure Login
- Forgot Password
- OTP Email Verification
- Account Lock after failed attempts

Password Rules

Current requirement:

- 6-digit password minimum

Recommended:

- Strong password policy
 - Combination of:
 - Numbers
 - Uppercase
 - Lowercase
 - Symbols
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5.2 Land Management

Features

- Unique Land ID
- Land mapping
- Customer-land relation
- Owner-land relation

- Worker-land assignment

Search System

Search by:

- Land ID
 - Customer Name
 - Owner Name
 - Worker Name
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5.3 Task Management

Task Features

- Create tasks
- Assign tasks
- Set deadlines
- Attach files
- Task status tracking

Task Status

- Pending
 - Started
 - In Progress
 - Completed
 - Delayed
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5.4 Media Management

Supported Uploads

Images

- JPG
- PNG
- WEBP

Videos

- MP4

- MOV

Documents

- PDF

Media Features

- Multiple upload support
 - Fast CDN delivery
 - Download option
 - Preview mode
 - Compression
 - Watermark support (optional)
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5.5 Communication System

Features

- Comment section
- Complaint system
- Notifications
- Task discussions

Communication Flow

Customer ↔ Owner

Owner ↔ Worker

Admin ↔ All Users

5.6 Reporting System

Reports

- Daily reports
- Land activity reports
- Worker performance reports
- Task completion reports

Export Options

- PDF

- Excel
 - CSV
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6. Non-Functional Requirements

6.1 Performance

Requirements

- API response < 1 second target
- Media load optimization
- CDN integration
- Efficient caching

Scaling

- Horizontal scaling support
 - Load balancing
 - Queue processing
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6.2 Security

Security Layers

- JWT Authentication
 - RBAC
 - SQL Injection protection
 - XSS protection
 - CSRF protection
 - API rate limiting
 - Encrypted storage
 - Secure file uploads
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6.3 Reliability

Requirements

- 24/7 support architecture

- Automatic backup
 - Crash recovery
 - Monitoring system
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6.4 Multi-Tenant Architecture

Tenant Isolation

Each tenant/company data must be isolated:

- Separate land access
 - Separate user access
 - Secure tenant validation
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7. Suggested System Architecture

Frontend Layer

React JS

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API Gateway

Backend Layer

FastAPI Services:

- Auth Service
- Task Service
- Media Service
- Notification Service
- Report Service

Database Layer

PostgreSQL

Storage

- Local Storage (Phase 1)
 - AWS S3 / Cloudflare R2 (Phase 2)
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8. Suggested Database Tables

Core Tables

- users
 - roles
 - lands
 - tasks
 - task_media
 - complaints
 - comments
 - login_logs
 - notifications
 - reports
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9. UI/UX Requirements

Dashboard Style

- Premium Agri Dashboard
- Glassmorphism effects
- Smooth animations
- Fast transitions

Mobile Experience

- One-hand usage support
 - Responsive cards
 - Offline caching (future)
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10. API Requirements

API Standards

- RESTful APIs
- Swagger Documentation
- Versioning support

Main APIs

- Authentication APIs

- Task APIs
 - Media APIs
 - Reporting APIs
 - Search APIs
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11. Future Enhancements

Phase 2 Features

- AI crop analysis
 - Drone integration
 - Satellite monitoring
 - Weather API integration
 - GPS tracking
 - Voice support in Tamil/English
 - Mobile App (React Native)
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12. Deployment Recommendations

Recommended Deployment

Frontend

- Vercel / Netlify

Backend

- Docker + Ubuntu Server

Database

- PostgreSQL Dedicated Server

Reverse Proxy

- NGINX
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13. Success Metrics

KPIs

- Upload success rate
 - System uptime
 - Task completion rate
 - User satisfaction
 - API response time
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14. Risks & Mitigation

Risk	Solution
Large video uploads	Compression & chunk upload
Slow database queries	Index optimization
Unauthorized access	RBAC + JWT
High traffic	Load balancing
Data loss	Automated backups

15. Final Product Vision

A premium enterprise agriculture management ecosystem where:

- Land owners can remotely monitor all farming activities.
- Workers provide transparent proof-based updates.
- Owners manage farm operations efficiently.
- Admin controls the entire ecosystem securely.
- Platform scales to enterprise agricultural businesses across India.